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PRN: B25ME1027 Div: SY3 Class: C3

INPUT:

```
#include <iostream>
using namespace std;

class Complex
{
 public:

 int real,img;

 void accept(int r, int i)
  { real=r;img=i; }

 void display()
 {
  cout<<"";
  if (img>=0)
   cout<<real<<"+i"<<img;
  else
   cout<<real<<"-i"<<img;
 }


Complex add(Complex c)
    {
    Complex x;
    x.real = real + c.real;
    x.img = img + c.img;
    return x;
    }

 Complex sub(Complex c)
    {
    Complex x;
    x.real = real - c.real;
    x.img = img - c.img;
```

```
    return x;
    }
 Complex mul(Complex c)
    {
    Complex x;
    x.real = real * c.real - img * c.img;
    x.img  = real * c.img + img * c.real;
    return x;
    }

    Complex div(Complex c)
    {
    Complex x;
    double denom = c.real * c.real + c.img * c.img;

    x.real = (real * c.real + img * c.img) / denom;
    x.img  = (img * c.real - real * c.img) / denom;

    return x;
    }


};

int main()
  {
    Complex c1, c2, sum, diff, product, quotient;

  c1.accept(3, 5);
  c2.accept(5, 2);

  cout << "\nFirst Complex Number: ";
  c1.display();

  cout << "\nSecond Complex Number: ";
  c2.display();

  sum = c1.add(c2);
  cout << "\nAddition of the complex numbers is: ";
  sum.display();
```

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  diff = c1.sub(c2);
  cout << "\nSubtraction of these complex numbers is: ";
  diff.display();

  product = c1.mul(c2);
  cout << "\nMultiplication: ";
  product.display();

  quotient = c1.div(c2);
  cout << "\nDivision: ";
  quotient.display();

    cout << endl;

    return 0;
}
```

OUTPUT:

